

DEVA ANAND

devaanand@umass.edu • (413) 315-1715 • devaanand.com • linkedin.com/in/devaaa • github.com/Deva-1903

Education

MS, Computer Science, University of Massachusetts Amherst 2025–2027

Advanced NLP, Advanced Machine Learning, Optimization in CS, Systems for Data Science, Performance Engineering

BE, Computer Science and Engineering, Vels Institute of Science, Technology and Advanced Studies 2019–2023

Professional Experience

Culture & Morality Lab, UMass Amherst, Lab Assistant | *Amherst, MA* Jul 2026 – Present

- Sole engineer building the lab's CCR text-analysis platform (Python, **FastAPI**, React) under full technical ownership, designing a DB-persisted job queue with orphaned-job recovery to keep long-running analysis jobs reliable.
- Built a structured data-quality warning system (8 machine-readable codes) and a versioned construct library (**99 constructs from 38 questionnaires**); grew the backend suite to **40 tests** (hermetic CI).

Wysa, Backend Engineer | *Bengaluru, India* Jul 2023 – Jul 2025

- Engineered a multi-tenant NHS eTriage backend (Node.js, MongoDB) for **8+ UK clinical clients**, processing **10,000+ monthly triage submissions** in a mission-critical clinical workflow via Mayden iaptus REST integrations; contributed to **\$340K+** in revenue.
- Architected a full-stack ML training-data annotation platform (React, Node.js, MongoDB) used daily by the AI team, cutting manual annotation by **100+ hours/month** and lifting labeling throughput by **60%**.
- Owned backend security and reliability for the eTriage stack, implementing **AWS KMS** encryption for clinical PII, hardening authentication and rate limiting, and remediating **20+ VAPT findings** with regression-tested fixes.
- Diagnosed and resolved production incidents across tenants, monitoring logs during releases and client integrations and shipping targeted fixes for tenant-specific issues to keep the multi-tenant eTriage service stable.

Cario Growth Services, Backend Developer Intern | *Chennai, India* Nov 2022 – May 2023

- Designed and shipped **20+ RESTful APIs** (Node.js, Fastify, PostgreSQL) with input validation and clear, backward-compatible contracts consumed in production by the frontend team.
- Migrated core repositories from JavaScript to TypeScript and shipped a production LLM feature (**4-bit quantized Llama-7B via llama.cpp**) with async queued generation, rate limits, and content-safety filtering.

Projects

AgenticSearch – Provenance-First Entity Discovery (GitHub | Demo) Python, FastAPI, OpenAI / Groq, SQLite

- Built a backend service exposing **REST API** endpoints with typed request planning, deterministic extraction with LLM fallback, and provider routing/fallback; hardened with **150+ automated tests** to keep behavior reliable under changing inputs.

cf-ai-research-scout – Edge-Native AI Assistant (GitHub) TypeScript, Cloudflare Workers, Durable Objects, D1

- Built a distributed, **event-driven** service with durable multi-step workflows that survive process restarts and tool failures, streaming results to the client over WebSocket in **real time**.

Spark ETL Performance Optimization (GitHub) PySpark, Spark UI, SQL, NYC TLC Dataset

- Optimized a high-volume data pipeline over a **~1.4B-row** dataset with broadcast joins, AQE, partitioning, and skew handling, cutting end-to-end runtime **~25%** and tail-latency ratio **2.14x to 1.74x**, profiled via Spark UI.

Publications

- "Alzheimer's Disease Classification using Transfer Learning," IEEE CONIT 2023, first author, deep transfer learning on neuroimaging data. (Paper)

Skills

Languages: Java, C++, Python, TypeScript, JavaScript, SQL, Bash

Backend & APIs: REST APIs, Node.js, React, Next.js, FastAPI, object-oriented design (SOLID), event-driven systems, messaging (Kafka), caching (Redis), concurrency

Databases: PostgreSQL, MongoDB, Redis, SQLite

Testing & DevOps: Integration testing, automated testing, CI/CD, Docker, Kubernetes, Git, GitHub Actions

Cloud / Platforms: AWS, GCP, Azure, Linux, MongoDB Atlas

Open Source & Community

- Built **Underdogs Fitness** (Website), a production MERN platform, solo as a freelance engagement; live **3+ years**. - Selected for **GirlScript Summer of Code 2023**; contributed to Linkfree, Freehit, and ProjectsHut. - Tutored **30+ underprivileged students** in programming at Sayur, a non-profit in Tamil Nadu.